

1 Question 1

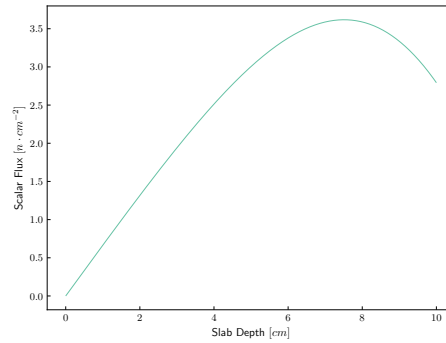
To begin, the angular flux is:

$$\psi(x, \mu) = \sin\left(\frac{x}{4}\right) \left[1 + \left(\mu - \frac{1}{10}x \right)^2 \right]$$

1.1 Part A

The scalar flux is the integral with respect to angle

$$\begin{aligned} \phi(x) &= \int_{-1}^1 d\mu \sin\left(\frac{x}{4}\right) \left[1 + \left(\mu - \frac{1}{10}x \right)^2 \right] \\ &= \sin\left(\frac{x}{4}\right) \int_{-1}^1 d\mu \left[1 + \left(\mu - \frac{1}{10}x \right)^2 \right] \\ &= \sin\left(\frac{x}{4}\right) \left(\frac{8}{3} + \frac{x^2}{50} \right) \end{aligned}$$



1.2 Part B

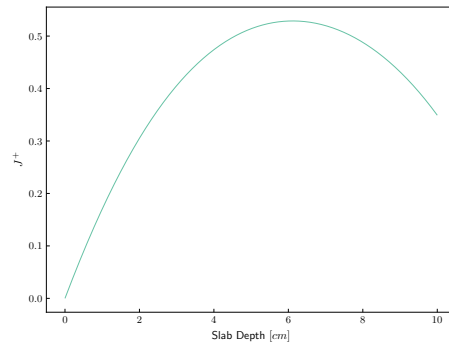
The total neutrons are found through integrating the scalar flux in space

$$\begin{aligned} N &= \int_0^{10} dx \sin\left(\frac{x}{4}\right) \left(\frac{8}{3} + \frac{x^2}{50} \right) \\ &= 24.8406415566746 \end{aligned}$$

1.3 Part C

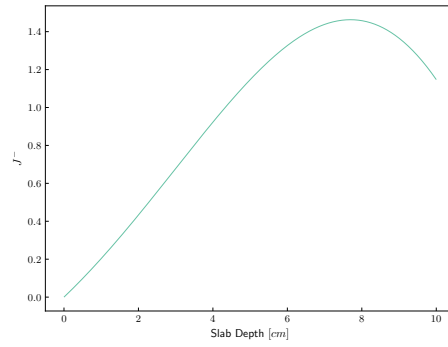
First, J^+

$$\begin{aligned}
 J^+(x) &= \int_0^1 d\mu \, \mu \psi(x, \mu) \\
 &= \int_0^1 d\mu \, \mu \sin\left(\frac{x}{4}\right) \left[1 + \left(\mu - \frac{1}{10}x \right)^2 \right] \\
 &= \sin\left(\frac{x}{4}\right) \left(\frac{x^2}{200} - \frac{x}{15} + \frac{3}{4} \right)
 \end{aligned}$$



Then, J^-

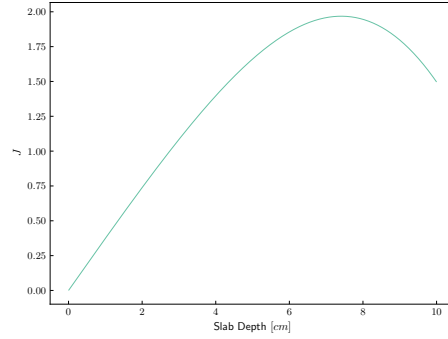
$$\begin{aligned}
 J^-(x) &= - \int_- 1^0 d\mu \mu \psi(x, \mu) \\
 &= - \int_- 1^0 d\mu \mu \sin\left(\frac{x}{4}\right) \left[1 + \left(\mu - \frac{1}{10}x \right)^2 \right] \\
 &= \sin\left(\frac{x}{4}\right) \left(\frac{x^2}{200} + \frac{x}{15} + \frac{3}{4} \right)
 \end{aligned}$$



1.4 Part D

Particle current is defined via:

$$\begin{aligned}
 J(x) &= \int_{-}^{+} 1^1 d\mu \mu \psi(x, \mu) \\
 &= \int_{-}^{+} 1^1 d\mu \mu \sin\left(\frac{x}{4}\right) \left[1 + \left(\mu - \frac{1}{10}x\right)^2\right] \\
 &= -\sin\left(\frac{x}{4}\right) \cdot \frac{2x}{15}
 \end{aligned}$$



And the sum of the two components is:

$$\begin{aligned}
 J(x) &= J^+ - J^- \\
 &= \sin\left(\frac{x}{4}\right) \left(\frac{x^2}{200} - \frac{x}{15} + \frac{3}{4}\right) - \sin\left(\frac{x}{4}\right) \left(\frac{x^2}{200} + \frac{x}{15} + \frac{3}{4}\right) \\
 &= \sin\left(\frac{x}{4}\right) \cdot \frac{-2x}{15}
 \end{aligned}$$

1.5 Part E

$$\begin{aligned}
 N_{leak} &= J^-(0) + J^+(10) \\
 &= 0 + 0.34910875072730796 \\
 &= 0.34910875072730796
 \end{aligned}$$

1.6 Part F

$$\begin{aligned}
 dose &= \int_0^4 dx \sigma_{d,1} \phi(x) + \int_4^1 0 dx \sigma_{d,2} \phi(x) \\
 &= 25.94597371679638 + 137.5601276932073 \\
 &= 163.50610141
 \end{aligned}$$

2 Question 2

To begin, the transport equation for this question can be simplified to:

$$\begin{aligned}\mu \frac{\partial \psi_1}{\partial x} &= -\sigma_{t,1} \psi_1 \\ \mu \frac{\partial \psi_2}{\partial x} &= -\sigma_{t,2} \psi_2\end{aligned}$$

with the following boundary conditions

$$\begin{aligned}\psi_1(0) &= \psi_0 \\ \psi_1(a) &= \psi_2(a) \\ j_1(a) &= j_2(b) \\ \psi_2(\infty) &= 0\end{aligned}$$

So then the general solutions are

$$\begin{aligned}\psi_1(x) &= \psi_0 e^{-\sigma_{t,1} x} \\ \psi_2(x) &= \psi_0 e^{-\sigma_{t,1} a} e^{-\sigma_{t,2}(x-a)}\end{aligned}$$

Then the mean free path is the first moment of the pdf:

$$\begin{aligned}mfp &= \int_0^\infty dx x [\sigma_{t,1} \psi_1(x) + \sigma_{t,2} \psi_2(x)] \\ &= \int_0^a dx x \sigma_{t,1} \psi_1(x) + \int_a^\infty dx x \sigma_{t,2} \psi_2(x) \\ &= \int_0^a dx x \sigma_{t,1} e^{-\sigma_{t,1} x} + \int_a^\infty dx x \sigma_{t,2} e^{-\sigma_{t,1} a} e^{-\sigma_{t,2}(x-a)} \\ &= \frac{e^{-a\sigma_{t,1}} \left(\frac{\sigma_{t,1}}{\sigma_{t,2}} - 1 \right) + 1}{\sigma_{t,1}}\end{aligned}$$

Then, plugging in an infinite optical thickness for region 1, we obtain

$$\begin{aligned}mfp &= \frac{e^{-\infty} \left(\frac{\sigma_{t,1}}{\sigma_{t,2}} - 1 \right) + 1}{\sigma_{t,1}} \\ &= \frac{1}{\sigma_{t,1}}\end{aligned}$$

Then, plugging in $\sigma_{t,2} = \sigma_{t,1}$:

$$\begin{aligned}&= \frac{e^{-a\sigma_{t,1}} (1 - 1) + 1}{\sigma_{t,1}} \\ &= \frac{1}{\sigma_{t,1}}\end{aligned}$$

3 Question 3

4 Question 4

4.1 Part A

$$\begin{aligned}
& \frac{1}{c} \frac{\partial \psi^\gamma(\vec{r}, \hat{\Omega}, E, t)}{\partial t} + \hat{\Omega} \cdot \nabla \psi^\gamma(\vec{r}, \hat{\Omega}, E, t) + \sigma_t^\gamma \psi^\gamma \\
&= \int_{4\pi} \int_0^\infty dE' d\hat{\Omega}' \sigma_s^\gamma(\vec{r}, \hat{\Omega}' \rightarrow \hat{\Omega}, E' \rightarrow E) \psi^\gamma(\vec{r}, \hat{\Omega}', E', t) \\
&\quad + \frac{\delta(E - m_e c^2)}{2\pi} \int_0^\infty \sigma_p^\gamma(\vec{r}, E) \psi(\vec{r}, \hat{\Omega}, E, t) + q^\gamma(\vec{r}, \hat{\Omega}, E, t)
\end{aligned}$$

4.2 Part B

$$\begin{aligned}
& \frac{1}{c} \frac{\partial \psi^\gamma(\vec{r}, \hat{\Omega}, E, t)}{\partial t} + \hat{\Omega} \cdot \nabla \psi^\gamma(\vec{r}, \hat{\Omega}, E, t) + \sigma_t^\gamma \psi^\gamma \\
&= \int_{4\pi} \int_0^\infty dE' d\hat{\Omega}' \sigma_s^\gamma(\vec{r}, \hat{\Omega}' \rightarrow \hat{\Omega}, E' \rightarrow E) \psi^n(\vec{r}, \hat{\Omega}', E', t) \\
&\quad + \frac{\delta(E - m_e c^2)}{2\pi} \int_0^\infty \sigma_p^\gamma(\vec{r}, E) \psi(\vec{r}, \hat{\Omega}, E, t) \\
&+ \frac{\chi^\gamma(\vec{r}, E' \rightarrow E)}{4\pi} \int_0^\infty dE' \nu^\gamma(\vec{r}, E) \sigma_t^n(\vec{r}, E) \psi^n(\vec{r}, \hat{\Omega}, E', t) + q^\gamma(\vec{r}, \hat{\Omega}, E, t)
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{v^n(E)} \frac{\partial \psi^n(\vec{r}, \hat{\Omega}, E, t)}{\partial t} + \hat{\Omega} \cdot \nabla \psi^n(\vec{r}, \hat{\Omega}, E, t) + \sigma_t^n \psi^n \\
&= \int_{4\pi} \int_0^\infty dE' d\hat{\Omega}' \sigma_{\gamma, n}^\gamma(\vec{r}, \hat{\Omega}' \rightarrow \hat{\Omega}, E' \rightarrow E) \psi^\gamma(\vec{r}, \hat{\Omega}', E', t) \\
&\quad + \int_{4\pi} \int_0^\infty dE' d\hat{\Omega}' \sigma_s^n(\vec{r}, \hat{\Omega}' \rightarrow \hat{\Omega}, E' \rightarrow E) \psi^n(\vec{r}, \hat{\Omega}', E', t) \\
&+ \frac{\chi^n(\vec{r}, E' \rightarrow E)}{4\pi} \int_0^\infty dE' \nu^n(\vec{r}, E) \sigma_f^n(\vec{r}, E) \psi^n(\vec{r}, \hat{\Omega}, E', t) + q^n(\vec{r}, \hat{\Omega}, E, t)
\end{aligned}$$